Department of Anaesthesia Quality Improvement Projects & Audits Report

Ysbyty Gwynedd

April 2016-2018

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Introduction

This report summarises clinical audit activity for the period 1st April 2016 to 31st March 2018 of Ysbyty Gwynedd Anaesthetic Department together with clinical audit activity registered from April 2018 onwards.

Since 1st April 2016, **19** QI projects have been completed and presented at local audit departmental meetings with a further **11** projects ongoing.

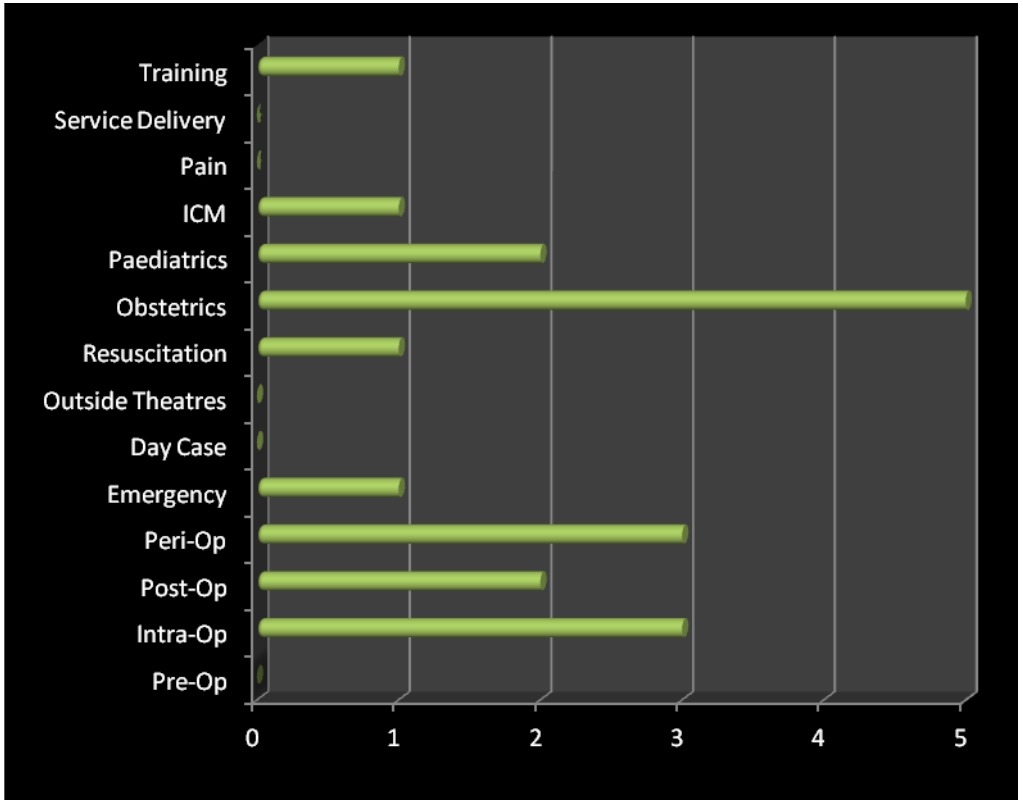


Figure 1: Completed Audits Per Primary Domain (N=19)

National Audit Involvement

The anaesthetic department had been involved in the following national audits.

- **National Emergency Laparotomy Audit (NELA)**
- **National Obstetric Anaesthetic Database (NOAD)**
- **NAP 6**
- **Intensive Care National Audit & Research Centre (ICNARC)**
- **TARN**
- **All Wales Post-partum Haemorrhage Collaboration (Obs Cymru)**
- **National Paediatric Audits**

Completed Projects

Audit Title	Domain	Lead/Authors	Period	Date Presented	Outcome
Introduction of Rectus Sheath Catheter Utilisation	Post-Op	S Clements	2016	N/A	Anecdotally poor administration of wound catheter boluses on ward. **Aim:** To provide access to elastomeric pumps and guidance on use of rectus sheath catheters. **Action:** Creation of hospital guidelines including In-Theatre Quick Guide. Provision of elastomeric pumps by pharmacy upon request. Purchase elastomeric pumps by theatres for out of hours use. **Future:** Consider provision of elastomeric pumps by Aseptic dept. for weekend use (storage in recovery fridge)
Anaesthetic Record Completion Audit	Intra-Op	S Beckett E F-Hills	2016	14th June 2016	**Aim:** Assess Anaesthetists compliance with GMC and AAGBI standards **Results:** Name of surgeon 62%, Operation performed 72% Evidence of risk discussion: 60%,Blood loss 28% **Recommendations:** Plan to revise chart to promote documentation.
Trauma Theatre Utilisation Audit	Peri-Op	S Patil P Kainth L Handcock	15 Dec 2015- 15 March 2016	14th June 2016	**Aim:** Improve start times to improve theatre utilisation. Results: Start >9:45: 19.8% overall. 43% weekends. Reason rarely recorded, but 50% Anaesthetist. Mean sent-arrival time 7mins. **Recommendations:** Improve reasons for delay reporting to determine target for improvement.

Availability of airway equipment to carry out a rapid sequence induction in ED Resus	Emergency	B Ellis	2016	14th June 2016	**Aim:** To quantitively assess the availability of airway equipment in the resus room of the emergency department according to local current guidelines. **Results:** Poor adherence to local policy with missing airway equipment & lack of standardization eg. Bougie 0%, 2nd laryngoscope 0%, HME/Cath mount: 45% **Recommendations:** Develop & Introduce RSI kit bag sealed and checked.
Standardising Anaesthetic Airway & CICO equipment	Intra-Op	I Rees C Dean S Clements D Maloney	2016	14th June 2016	**Aim:** Pre-implementation audit of Plan D equipment availability & Survey of standard airway trolleys **Results:** No standardisation of trolley type, no equipment checklist, airway cart used to store non airway items. 5/12 theatres had Plan D equipment only. **Outcomes:** New standardised Airway carts procured & all theatres equipped with CICO equipment
Perioperative Management of Diabetes	Peri-Op	K Braz A Nagy S Mitra	2016	14th June 2016	**Aim:** Review 2015 AAGBI guidelines with staff survey. **Results:** Poor knowledge of best practice DM management. **Recommendations**: Raise awareness of recent guideline publication. Send out new guidelines to doctors. Re-audit 6 months.
On Call Vascular Access Request	Peri-Op	M Marston	Oct 2016	13th Jan 2017	**Aim:**To determine frequency & reason for requesting vascular access by on call anaesthetist **Recommendations:** There is a definite need to introduce intermediate PVC (Midline) provision and develop local guidelines on appropriate options for long term vascular access.
Obstetric Anaesthesia Services Audit 2016	Obstetrics	S Patil G Roberts	2016	13th March 2017	Review of workload and complications relative to national data **Results:** GA LSCS rate: YG 11.7% (Nat 7.6%) lab analgesia 500. High spinals:3, Blood patches:8.

					Epidurals: 88% effective, 82% patient satisfaction.
Audit of Post Regional Anaesthesia Ward Care	Obstetrics	A Patel G Roberts	Feb 2017	13th March 2017	**Aim:** To determine whether post operative ward observations meet guidance standards following obstetric regional anaesthesia **Standard:** OAA: documented evidence of hrly vita signs for 12hrs & motor block assessment until full recovery & written info received upon discharge. **Results:** 0% documentation for motor block. 0% had hrly vitals documented. No women received written documentation. **Outcomes:** Introduce new obstetric obs chart, Implement vital sign reminder poster & written info sheet created.
Re-Audit of Theatre Exit Checks	Post-Op	K Dhar S Clements	May 2017	N/A	In 2015, a snapshot of anaesthetic staff practice found a deficiency in carrying out certain key tasks which impact on patient safety, prior to patients leaving the operating theatre. Introduction of a simple, rapid, theatre exit check (YODEL) had a positive influence on task completion. **Aim:** To check ongoing compliance with YODEL check: Machine on standby, Trolley Oxygen flow check, Monitor case reset, Used drugs discarded. **Method:** 2 day prospective Anaesthetist practice observation **Result:** All 4 standards achieved in 81% of cases. **Action:** Reaudit in 1 Year
Anaesthetic Record Keeping Legibility Audit	Intra-Op	M Marston T Lloyd S Mitra	2017	3rd Aug 2017	**Aim:** Assess Anaesthetists compliance with GMC and AAGBI standards **Standards:** All drug entries should be legible, abbreviations only if there is an official hospital glossary. Dose & time. **Method:** 50 chart review. **Results:** 51% legible (2016 audit: 49%) **Actions:**

					• Ensure department makes rigorous efforts to ensure legibility • Only acceptable abbreviations to be used – 'can another medical professional read and understand the notes?' • Re-audit in 3 months • Contact MDU/MPS for medicolegal talk
CVC Insertion – A Quality Improvement Project	ITU	K Mottart L Roberts J Rochester	2016-2017	12th September 2017	**Aim**: Improve CVC documentation in line with local then national guidance **Action:** Update and improved awareness of CVC insertion sticker **Results:** Compliance with minimum standard documentation increased from 60% to 75%. **Future:** Identify other factors to reduce sepsis associated with CVC insertion. Create All Wales CVC insertion sticker
Trauma Times to CT (STAG audit)	Resuscitation	M knights	2017	8th Dec 2017	**Aim:** To review Time to CT for major trauma patients. Targets: CT within 30mins. MTC within 2 hrs of arrival. **Results:** Median age 63. Mech: Fall most common. (<2M 2nd most common). Time from arrival to CT 91mins. (Fastest 27mins). Time to MTC from arrival :5hrs 19 median: CT to decision:1 hr 50, Decision to leaving: 1hr 36. Over 70s had greater ISS but slow time to CT 145 v 48mins. **Conclusion:** Elderly patients are being under-triaged. **Recommendations:** Review trauma team call out. Early consultant input for potential transfers. Introduction of Pre-transfer and Neuro protection check list.
Airway Training Needs	Training	S Tobin L Connolly M Marston D Maloney	2017	23rd Jan 2018	**Aim:** Assess extent of departmental airway training **Standard:** NAP4 recommended regular formal training **Method:** Kit & training familiarity survey **Results:** 20% require training on Ambu scopes. 60% for Bonfils,

		S Clements J Walker			18% Aintree catheter. CICO: 81% trained in scalpel bougie. 79% trained in cannula technique, 67% Rapid O2, 69% cannula/Melker conversion. Courses: 81% attended CICO course. **Action:** Continue annual airway workshop.
Obstetric Anaesthesia Services Audit 2017	Obstetrics	D O-Donovan G Roberts	2017	14th March 2018	Review of workload and complications relative to national data **Results:** GA LSCS rate: YG 10.8% (Nat 7.6%), LSCS rate 26% YG (26% Nat). Lab analgesia: 496. High spinals: 0, Dural punctures: 2. No major complications. Blood pateches:10
Re-Audit of Post Regional Anaesthesia Ward Care	Obstetrics	B Ellis A Patel G Roberts	July 2017-Aug 2017	14th March 2018	**Aim:** To determine whether post operative ward observations meet guidance standards following obstetric regional anaesthesia following previous audit. **Standard:** OAA: documented evidence of hrly vita signs for 12hrs & motor block assessment until full recovery & written info received upon discharge **Results:** motor assessment: 100%, Vital signs: Variability beyond 1st 2 hrs.
Audit of blood transfusion for patients with postpartum haemorrhage following LSCS	Obstetrics	T Shambrook E Ardelean	1st June 2017 - 31st July 2017	14th March 2018	**Aim:** To assess if we are transfusing patients undergoing LSCS appropriately. **Standard:** Target Hb >80 g/l and asymptomatic. 100% of Major PPH have: POC, FBC/coag, Lactate, TXA. **Results:** Asymptomatic: Minor PPH=72%. Major PPH=10% Hb >80g/dl: Minor: 98%, Major=80%. pRBC. Hb>80g/L: 95%, (Major PPH:80%) Major PPH (N=10): ROTEM 70%, POC Hb 70%, FBC/Coag 60%, Lactate: 30% , TXA: 50%. **Recommendations:** Incorporate Obs Cymru into Obstetric induction for new anaesthetic staff.

National PAPAYA Audit Phase 1	Paediatrics	L Warnock L Roberts	March 2017	TBP	**Retrospective review:** Unplanned admission after day case procedures In paediatric patients 0-16 years **Aim:** Identify number of paediatric day case procedures and identify number of unplanned admissions – via IT data. **Action:** 1320 day cases identified from IT, 130 identified as unplanned admissions. **Future:** Improve data accuracy as number of unplanned admissions seems inappropriately high.
National PAPAYA Audit Phase 2	Paediatrics	L Warnock L Roberts	October 2017 – December 2017	TBP	**Aims:** 1. To more accurately identify the proportion of unplanned admissions after paediatric day case anaesthesia across the UK in both specialist and non-specialist paediatric centres in a national prospective audit. 2. To identify common contributors involved in unplanned admissions. 3. To generate data to target future quality improvement projects by the APA, PATRN and individual centres, and hence improve day case anaesthesia for children. **Results:** 6/25 admissions were unplanned (3%), 50% for pain, 30% PONV, 16% Urinary retention, 16% physio review. 96% of patients that had unplanned admission had an anaesthetic related reason that contributed to their admission . **Future:** Consider local agreed guide for pain / PONV for paediatric cases.

Ongoing Projects (to date)

Audit Title	Domain	Lead/Authors	Period	Planned Presentation Date	Objectives
Post Op Plan Documentation	Post-Op	Dr S Clements	2018	12th June 2018	Anaesthetic documentation of a post-op plan is good practice for continuity of care and for medico legal reasons. Aim: For a post op plan to be documented in 100% of cases
Quality of anaesthetic chart documentation	Peri-op	Dr E Farley-Hills	?	12th June 2018	To assess improvement in anaesthetic documentation with the introduction of a new chart, based on AAGBI guidelines.
Guidelines for Perioperative Management of Obstructive Sleep Apnoea	Peri-op	Dr L Connolly Dr S Mitra	2017-2018	12th June 2018	There is a 45% cancellation rate on day of surgery due to lack of HDU bed. To standardise approach to patients with OSA & reduce cancellation rates on the day of admission
Pre-operative test rationalisation	Pre-op	Dr Lloyd-Davies	October 2017	12th June 2018	To reduce over requests of preoperative tests in line with NICE guidance 2016.
Anaesthetic Abbreviations list	Peri-op	Dr D Ambike Dr J Awuku	January 2018	?12th June 2018	To create trust level " Anaesthetic Abbreviations List" which will be officially approved by anaesthetic department at Ysbyty Gwynedd for use in day to day practice.
Management of Reversal of Neuromuscular Blockade	Intra-op	Dr B Eagle, Dr D Maloney Dr S Clements	August 2017		Provide guidance on Reversal of Neuromuscular blockade To rationalise use of Suggamadex
Audit on perioperative management of adults with diabetes undergoing elective surgery	Peri-op	Dr E Ardelean Dr S Mitra	February 2018	12th June 2018	To assess if management of adults with diabetes undergoing elective procedures is in compliance with national guidance.

Audit of Preoperative Fasting	Pre-Op	Dr A Ninan Dr S Clements	February 2018	12th June 2018	To optimise current fasting times in line with national guidelines.
Re-audit of documentation of cocaine administration	Intra-Op	Dr H Williamson Dr D Maloney	March 2018	12th June 2018	To improve documentation of the administration of cocaine.
Audit of patients patient cancellations at out-patient clinics and interventions theatre lists for Pain Management services	Chronic Pain	Dr S Patil	March 2018	12th June 2018	identify the extent of the problem of non-attendance and unplanned cancellations for any reason.
Re-Auditing The Availability of Airway Equipment in The Emergency Department Following Introduction of New Airway Bags	Resus	Dr B Ellis, Dr M Knights Dr P Williams	March 2018	May 2018	To quantitatively assess the availability of airway equipment in the resus room of the emergency department according to the new local guidelines and compare to previous audit results of the old local guideline.
Prospective audit of peri-operative pain management in emergency Laparotomy patients at Ysbyty Gwynedd	Peri-Op	Dr C Dallimore Dr S Clements	April-May 2018	12th June 2018	Assess current Anaesthetic practice, patient pain scores in order to guide development of interventions to standardize and improve patient analgesia post operatively.

Way Forward

The Anaesthetic department as part of the wider Anaesthetic, Critical Care and Pain Managment CPG needs to agree priorities for clinical audit for 2018-19 based on patient safety, risks, complaints/litigation, local and national initiatives. These topics should form the audits for the department throughout the year.

The focus of clinical audit activity is to improve patient care. Presentations and audit meetings should concentrate on demonstrating this improvement through monitoring and aiding the implementation of change.

Outstanding audits to be finalised where possible.

National audits which this department plans to continue to be involved include NELA & NOAD. Ysbyty Gwynedd will continue to be participating in the All Wales post partum haemorrage collaboration.

Trainees will be be involved in projects coordinated by the Welsh Anaesthetic Audit Research and Engagement Network (WAAREN) overseen by the Research & Audit Federation of Trainees (RAFT).